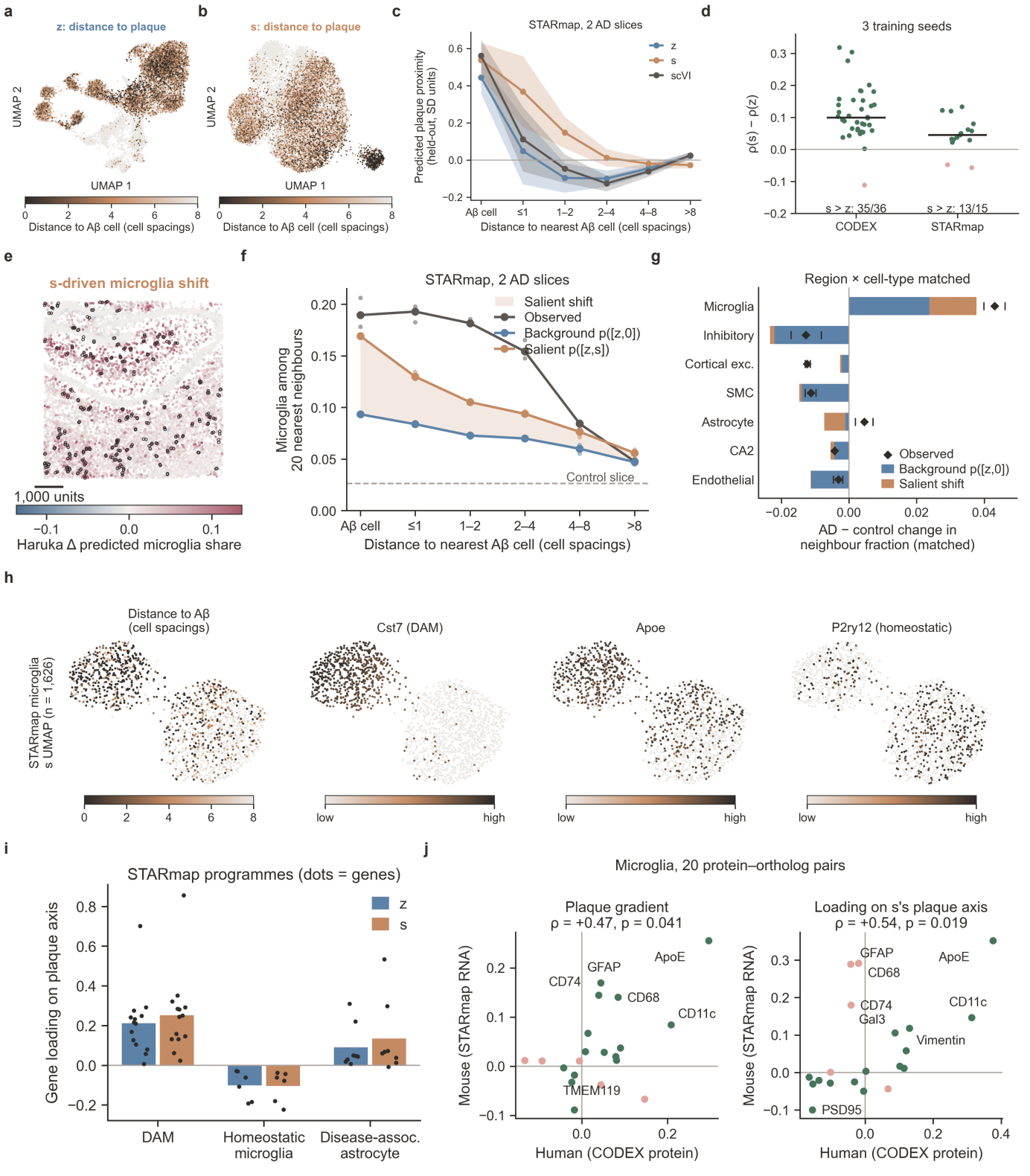

a
z: distance to plaque
UMAP 2
UMAP 1
0 2 4 6 8
Distance to Aβ cell (cell spacings)
b
s: distance to plaque
UMAP 2
UMAP 1
0 2 4 6 8
Distance to Aβ cell (cell spacings)
c
STARmap, 2 AD slices
Predicted plaque proximity (held-out, SD units)
z
s
scVI
Aβ cell ≤1 1–2 2–4 4–8 >8
Distance to nearest Aβ cell (cell spacings)
d
3 training seeds
ρ(s) − ρ(z)
s > z: 35/36
s > z: 13/15
CODEX
STARmap
e
s-driven microglia shift
1,000 units
−0.1 0.0 0.1
Haruka Δ predicted microglia share
f
STARmap, 2 AD slices
Microglia among 20 nearest neighbours
Salient shift
Observed
Background p([z,0])
Salient p([z,s])
Control slice
Aβ cell ≤1 1–2 2–4 4–8 >8
Distance to nearest Aβ cell (cell spacings)
g
Region × cell-type matched
Microglia
Inhibitory
Cortical exc.
SMC
Astrocyte
CA2
Endothelial
Observed
Background p([z,0])
Salient shift
AD − control change in neighbour fraction (matched)
h
Distance to Aβ (cell spacings)
Cst7 (DAM)
Apoe
P2ry12 (homeostatic)
STARmap microglia s UMAP (n = 1,626)
low high
i
STARmap programmes (dots = genes)
Gene loading on plaque axis
z
s
DAM
Homeostatic microglia
Disease-assoc. astrocyte
j
Microglia, 20 protein–ortholog pairs
Plaque gradient
ρ = +0.47, p = 0.041
Loading on s's plaque axis
ρ = +0.54, p = 0.019
Mouse (STARmap RNA)
Human (CODEX protein)
ApoE
GFAP
CD74
CD68
CD11c
TMEM119
Gal3
Vimentin
PSD95